

Functions and transformations - Set 1 - Solutions

Specification MM1.7, MM8.2, MM8.3, MM8.4.

Answer key

Question	Answer	Topic
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Worked solutions

Question 1 FUN-01 · Functions and transformations · spec MM8.2

The graph of $y = x^2$ is translated by 3 units in the positive x -direction and then stretched parallel to the y -axis with scale factor 2. What is the equation of the resulting graph?

Answer: A

Translation: $y = (x - 3)^2$. Stretch by factor 2 in the y -direction: $y = 2(x - 3)^2$.

Full method - how and why

Two transformations applied in a stated order. Apply them one at a time to the equation, in the order given.

- 1 Translate 3 in the positive x -direction: replace x by $x - 3$, giving $y = (x - 3)^2$.

why: A shift **right** by a replaces x with $x - a$ - the sign is opposite to what it looks like, because the new graph reaches a given height 3 later.

- 2 Stretch parallel to the y -axis with factor 2: multiply the whole function by 2, giving $y = 2(x - 3)^2$.

why: A vertical stretch multiplies every y -value, so it acts on the outside of the function.

Where it goes wrong

- Writing $(x + 3)^2$ - shifting right uses $x - 3$.
- Putting the 2 inside as $(2x - 3)^2$, which is a horizontal stretch, not a vertical one.

Takeaway Changes **inside** the function affect x and act in the opposite sense; changes **outside** affect y and act as they read.

Question 2 FUN-02 · Functions and transformations · spec MM8.2

$f(x) = x^2 - 4x$. The graph of $y = f(x + 2)$ has its minimum point at

Answer: A

$f(x) = (x - 2)^2 - 4$ has minimum at $(2, -4)$. Replacing x by $x + 2$ translates the graph 2 units to the left: $(0, -4)$.

Full method - how and why

Find the minimum of f first, then move it. Tracking the single turning point is quicker than re-completing the square.

- 1 Complete the square: $f(x) = x^2 - 4x = (x - 2)^2 - 4$, so f has its minimum at $(2, -4)$.

why: Half of -4 is -2 ; the -4 outside is the minimum value.

- 2 $y = f(x + 2)$ translates the graph 2 units to the **left**.

why: Replacing x by $x + a$ shifts left by a : the graph now reaches each value 2 earlier.

- 3 Move the turning point: $(2, -4) \rightarrow (0, -4)$.

why: Only the x -coordinate changes; a horizontal translation never alters y .

Where it goes wrong

- Shifting right and answering $(4, -4)$.
- Changing the y -coordinate as well, giving $(0, -2)$ or similar.

Takeaway $f(x + a)$ moves the graph a to the left; horizontal shifts leave y -values untouched.

Question 3 FUN-03 · Functions and transformations · spec MM8.2

Let $f(x) = 2x + 1$ and $g(x) = x^2 - 3$. Find $f(g(2)) - g(f(2))$.

Answer: C

$g(2) = 1$, $f(1) = 3$, $f(2) = 5$, $g(5) = 22$. So $3 - 22 = -19$.

Full method - how and why

Composite functions evaluated at a number. Work strictly from the inside out, and keep the two compositions separate.

1 $g(2) = 2^2 - 3 = 1$, then $f(1) = 2(1) + 1 = 3$. So $f(g(2)) = 3$.

why: In $f(g(2))$ the inner function acts first: feed 2 into g , then its output into f .

2 $f(2) = 2(2) + 1 = 5$, then $g(5) = 25 - 3 = 22$. So $g(f(2)) = 22$.

why: The other order gives a completely different value - composition is not commutative.

3 Subtract in the order asked: $3 - 22 = -19$.

why: The question is $f(g(2)) - g(f(2))$, so the small one comes first.

Where it goes wrong

- Computing $g(f(2)) - f(g(2))$ and answering 19.
- Applying the functions in the wrong order inside one composition.

Takeaway $f(g(x))$ means g first; check the order twice, because the wrong order is always an offered option.

Question 4 FUN-05 · Functions and transformations · spec MM8.2

The graph of $y = f(x)$ passes through (4, 6). Which point must lie on the graph of $y = 3f(2x) - 1$?

Answer: B

$f(2x)$ halves x -coordinates: (2, 6). Then $3f - 1$: $y = 18 - 1 = 17$. Point (2, 17).

Full method - how and why

Track a single known point through the transformations, remembering that inside changes act inversely on x .

1 Start with the known point (4, 6) on $y = f(x)$.

why: One point is all you are given, so the answer must be its image.

2 $f(2x)$ squashes horizontally by factor $\frac{1}{2}$: the point with x -coordinate 4 moves to $x = 2$, since $f(2 \times 2) = f(4) = 6$.

why: To get the same output from $f(2x)$ you need half the input - this is the inverse action of an inside change.

3 Multiply by 3: $y = 3 \times 6 = 18$. Then subtract 1: $y = 17$.

why: Outside changes act directly on the output, in the order written.

4 The point is (2, 17).

why: Check: $3f(2 \times 2) - 1 = 3(6) - 1 = 17$ ✓.

Where it goes wrong

- Doubling the x -coordinate to 8 instead of halving it.
- Forgetting the -1 and answering (2, 18).

Takeaway Substitute the point back into the new equation to check: $f(2x)$ at $x = 2$ really is $f(4)$.

Question 5 FUN-06 · Functions and transformations · spec MM8.2

The curve $y = x^2 - 6x + 5$ is reflected in the y -axis. Find the equation of the reflected curve.

Answer: A

Reflecting in the y -axis replaces x with $-x$: $y = x^2 + 6x + 5$.

Full method - how and why

Reflection in the y -axis replaces x by $-x$. Only the odd powers change sign.

- 1 Replace x with $-x$: $y = (-x)^2 - 6(-x) + 5$.

why: Reflecting in the y -axis maps the point (x, y) to $(-x, y)$, which is achieved by substituting $-x$.

- 2 Simplify: $(-x)^2 = x^2$ and $-6(-x) = +6x$, so $y = x^2 + 6x + 5$.

why: Even powers are unaffected; odd powers flip sign. The constant is also unchanged.

Where it goes wrong

- Changing the sign of the constant too, giving $x^2 + 6x - 5$.
- Confusing it with reflection in the x -axis, which would negate the whole right-hand side.

Takeaway y -axis reflection: $x \rightarrow -x$, so only odd-degree terms change sign. x -axis reflection: $y \rightarrow -y$, so everything does.

Question 6 FUN-08 · Functions and transformations · spec MM1.7

For which values of x is $\sqrt{x^2} = -x$?

Answer: C

$\sqrt{x^2} = |x|$, and $|x| = -x$ exactly when $x \leq 0$.

Full method - how and why

The key fact is $\sqrt{x^2} = |x|$, not x . The question is then about when $|x| = -x$.

- 1 Rewrite the left-hand side: $\sqrt{x^2} = |x|$.

why: The square-root symbol always means the **non-negative** root, so the output cannot be negative even when x is.

- 2 The equation becomes $|x| = -x$.

why: Both sides must therefore be ≥ 0 , which already forces $-x \geq 0$.

- 3 $|x| = -x$ exactly when $x \leq 0$ (for $x < 0$, $|x| = -x$; at $x = 0$ both sides are 0).

why: Split into cases $x \geq 0$ and $x < 0$, or recall the definition of $|x|$ directly.

Where it goes wrong

- Answering 'no real x ' on the grounds that a square root cannot equal a negative - but $-x$ is positive when x is negative.
- Excluding $x = 0$ and answering $x < 0$ only.

Takeaway $\sqrt{x^2} = |x|$; and $-x$ is positive when x is negative, so expressions with a minus sign can still be non-negative.

Question 7 FUN-09 · Functions and transformations · spec MM8.2

The graph of $y = \sin x$ is transformed to $y = 2\sin(3x) + 1$. What is the maximum value of the new function, and what is its period (in degrees)?

Answer: D

$\sin(3x)$ has period $360^\circ / 3 = 120^\circ$. $2\sin(3x)$ ranges from -2 to 2 , so $2\sin(3x) + 1$ has maximum 3 .

Full method - how and why

For $y = a\sin(bx) + c$, the amplitude, period and shift can be read straight off - as long as you know which letter does what.

- 1 The 3 inside compresses horizontally: period $= \frac{360^\circ}{3} = 120^\circ$.
why: $\sin(bx)$ completes a cycle when bx increases by 360° , i.e. when x increases by $360^\circ / b$.
- 2 The 2 outside is the amplitude: $2\sin(3x)$ ranges from -2 to 2 .
why: Multiplying the function stretches the range by that factor, symmetrically about zero.
- 3 The $+1$ raises everything: the range becomes -1 to 3 , so the maximum is 3 .
why: A vertical translation moves the whole wave, including its maximum.

Where it goes wrong

- Multiplying the period by 3 to get 1080° instead of dividing.
- Giving the amplitude 2 as the maximum and ignoring the $+1$.

Takeaway In $a\sin(bx) + c$: amplitude $|a|$, period $360^\circ / b$, midline $y = c$, maximum $c + |a|$.

Question 8 FUN-11 · Functions and transformations · spec MM8.2

The graph of $y = f(x)$ is transformed by the sequence: reflect in the x -axis, then translate 2 units in the positive y -direction, then reflect in the y -axis. The resulting graph has equation

Answer: D

Reflect in x -axis: $y = -f(x)$. Translate up 2: $y = -f(x) + 2$. Reflect in y -axis (replace x by $-x$): $y = 2 - f(-x)$.

Full method - how and why

Three transformations in a stated order. Apply them to the equation one at a time, and be careful that the last one acts on everything built so far.

- 1 Reflect in the x -axis: $y = -f(x)$.
why: Negating the output flips the graph vertically.
- 2 Translate 2 up: $y = -f(x) + 2$.
why: Add to the whole expression, not just to f .
- 3 Reflect in the y -axis: replace **every** x by $-x$, giving $y = -f(-x) + 2 = 2 - f(-x)$.
why: Only x is replaced; the constant 2 is unaffected because it does not involve x .

Where it goes wrong

- Writing $-f(-x) - 2$ by also negating the constant.
- Applying the transformations in the wrong order - with mixed inside/outside operations, order matters.

Takeaway Rewrite the equation after each step; a y -axis reflection substitutes $-x$ for x everywhere it appears.

Question 9 FUN-13 · Functions and transformations · spec MM8.4

The graph of $y = a(x + b)^2 + c$ has vertex $(-2, 5)$ and passes through $(0, -3)$. Find a .

Answer: A

$b = 2, c = 5$. At $x = 0$: $4a + 5 = -3$, so $a = -2$.

Full method - how and why

The completed-square form is handed to you, so the vertex gives two constants and the extra point gives the third.

- 1 Match the vertex: $y = a(x + b)^2 + c$ has vertex $(-b, c)$, and we are told it is $(-2, 5)$, so $b = 2$ and $c = 5$.

why: Note the sign: the form uses $x + b$, so the vertex x -coordinate is $-b$, not b .

- 2 Substitute the point $(0, -3)$: $a(0 + 2)^2 + 5 = -3$.

why: One point is enough to find the remaining unknown.

- 3 Solve: $4a = -8$, so $a = -2$.

why: Negative, which makes sense: the curve passes below its vertex, so it opens downwards.

Where it goes wrong

- Taking $b = -2$ and getting $a = -\frac{1}{2}$ from $a(0 - 2)^2 \dots$ (the same here, but the sign confusion bites when the point is not symmetric).
- Answering $+2$ and ignoring that the parabola must open downwards to reach $y = -3$.

Takeaway Sanity-check the sign of a against the picture: if the curve drops below the vertex, $a < 0$.

Question 10 FUN-14 · Functions and transformations · spec MM8.2

The graph of $y = f(x)$ has a maximum point at $(3, 8)$. The graph of $y = f(x - a) + b$ has its maximum at $(1, 2)$. Find $a + b$.

Answer: D

The maximum moves from $(3, 8)$ to $(3 + a, 8 + b) = (1, 2)$, so $a = -2, b = -6$ and $a + b = -8$.

Full method - how and why

Follow the turning point through the transformation and match coordinates - each coordinate gives one equation.

- 1 $y = f(x - a) + b$ translates the graph a right and b up, so the maximum moves from $(3, 8)$ to $(3 + a, 8 + b)$.

why: A translation moves every point, including the turning point, by the same vector.

- 2 Match the x -coordinates: $3 + a = 1$, so $a = -2$.

why: A negative a simply means the shift is to the left.

- 3 Match the y -coordinates: $8 + b = 2$, so $b = -6$.

why: Same for the vertical shift.

- 4 Add: $a + b = -8$.

why: The question asks for the sum, not for either value alone.

Where it goes wrong

- Using $3 - a = 1$ and getting $a = 2$, which flips the sign of the answer.
- Answering -2 or -6 (one of the two constants) instead of their sum.

Takeaway $f(x - a) + b$ shifts by $(+a, +b)$; set up one equation per coordinate.

Question 11 FUN-15 · Functions and transformations · spec MM8.2

Which of the following is the equation of the graph obtained by stretching $y = x^3 - 3x$ parallel to the x -axis with scale factor $\frac{1}{2}$?

Answer: E

A stretch parallel to the x -axis with factor $\frac{1}{2}$ gives $y = f(2x) = (2x)^3 - 3(2x) = 8x^3 - 6x$.

Full method - how and why

A horizontal stretch of factor $\frac{1}{2}$ replaces x by $2x$ - the reciprocal of the factor.

1 A stretch parallel to the x -axis with scale factor k gives $y = f(\frac{x}{k})$; with $k = \frac{1}{2}$ that is $y = f(2x)$.

why: Inside changes act inversely: to squash the graph to half its width you must double the input.

2 Substitute: $y = (2x)^3 - 3(2x)$.

why: Replace every x in the formula, including the one in the linear term.

3 Simplify: $8x^3 - 6x$.

why: $(2x)^3 = 8x^3$ - the coefficient is cubed too.

Where it goes wrong

- Replacing only the first x , giving $8x^3 - 3x$ - an offered option.
- Using $y = f(x/2)$ and getting $\frac{x^3}{8} - \frac{3x}{2}$, which is a stretch of factor 2.

Takeaway Stretch factor k parallel to the x -axis $\Rightarrow$ replace x by x/k ; substitute into **every** occurrence.

Question 12 FUN-17 · Functions and transformations · spec MM8.3

The line $y = mx + c$ passes through (2, 7). If m is increased by 1 and c is decreased by 2, the new line passes through (2, k). Find k .

Answer: B

New value at $x = 2$: $(m + 1) \cdot 2 + (c - 2) = 2m + c = 7$. So $k = 7$.

Full method - how and why

The two changes are designed to cancel at $x = 2$. Substitute and watch the terms collapse.

1 From the original line: $2m + c = 7$.

why: The line passes through (2, 7), so substituting gives one equation in m and c .

2 The new line is $y = (m + 1)x + (c - 2)$; evaluate it at $x = 2$: $2(m + 1) + (c - 2) = 2m + 2 + c - 2$.

why: Substitute $x = 2$ and expand - do not try to find m and c , which are not determined.

3 Simplify: $2m + c = 7$, so $k = 7$.

why: The +2 from the gradient change and the -2 from the intercept change cancel exactly.

Where it goes wrong

- Trying to solve for m and c individually - there is not enough information, and none is needed.
- Forgetting that the gradient change is multiplied by $x = 2$, and answering 5 or 9.

Takeaway When a question gives only one equation for two unknowns, look for a combination that is determined - here $2m + c$.